

1 Linear ODE theory and solving 1st order linear ODEs



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03

Introduction to Differential Equations

Instructor: David Earn

Lecture 7

Linear ODE theory and solving 1st order linear ODEs
Tuesday 22 September 2026

Announcements

- The TA is Illya Kierkosz (kierkosi@mcmaster.ca)
 - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: <https://davidearn.github.io/math2c/>
- The course outline is on the [Simple Syllabus](#) web site.
- [Avenue to Learn](#) provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on [childsmath](#).
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
 - Subject: Math 2C03: confusion about assignment 1
 - Subject: Math 2C03: conflict with test window
- Office hour: Thursdays, 9:30–10:20 am, after class.

Linear ODEs

Classification of ODEs by Linearity

The *generic linear ODE* of n^{th} order can be written

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x) y = g(x)$$

or, equivalently,

$$a_n(x) y^{(n)} + a_{n-1}(x) y^{(n-1)} + \cdots + a_1(x) y' + a_0(x) y = g(x)$$

Note: each *coefficient* a_i is a function **only** of the independent variable x , not a function of y , y' , ..., $y^{(n)}$. The function g on the RHS above is also a function **only** of x . The a_i 's and g *can be nonlinear functions of x* .

Let's reconsider **why** such ODEs (and **only** such ODEs) are called *linear*...

Classification of ODEs by Linearity

First, recall the notion of linearity in general:

- If V is a vector space and $F : V \rightarrow V$ then F is **linear** if

$$F(au + bv) = aF(u) + bF(v)$$

for any $u, v \in V$ and any $a, b \in \mathbb{R}$.

- Equivalently, F is **linear** if

$$\begin{aligned} F(au) &= aF(u) && \text{for any } a \in \mathbb{R}, \\ \text{and } F(u + v) &= F(u) + F(v) && \text{for any } u, v \in V. \end{aligned}$$

- If F is not linear, we say it is **nonlinear**.
- If F is a function of several variables, it can be linear in some variables, yet nonlinear in other variables.

Classification of ODEs by Linearity

Example (Linear vs nonlinear)

Consider $F(x, y) = x^2 y$.

Is F linear wrt x ?

i.e., Is $F(ax_1 + bx_2, y) = aF(x_1, y) + bF(x_2, y)$ for all $a, b \in \mathbb{R}$?

Check if true for **arbitrary** real constants a, b :

$$\begin{aligned} F(ax_1 + bx_2, y) &= (ax_1 + bx_2)^2 y = [(ax_1)^2 + 2abx_1x_2 + (bx_2)^2] y \\ &= (ax_1)^2 y + (bx_2)^2 y + (2abx_1x_2) y \\ &= a^2 F(x_1, y) + b^2 F(x_2, y) + (2abx_1x_2) y \\ &\neq a F(x_1, y) + b F(x_2, y) \quad \forall a, b \quad \implies \text{nonlinear in } x \end{aligned}$$

Is F linear wrt y ? Check:

$$\begin{aligned} F(x, ay_1 + by_2) &= x^2 (ay_1 + by_2) = ax^2 y_1 + bx^2 y_2 \\ &= a F(x, y_1) + b F(x, y_2) \quad \forall a, b \quad \implies \text{linear in } y \end{aligned}$$

Classification of ODEs by Linearity

Example (Linear vs nonlinear)

Consider $F(x, y, y') = x^2 \frac{dy}{dx}$.

Is F linear wrt x ? NO, as on previous slide. $\implies$ *nonlinear in x*

Is F linear wrt y ?

i.e., Is $F(x, ay_1 + by_2) = aF(x, y_1) + bF(x, y_2)$ for all $a, b \in \mathbb{R}$?

Check:

$$\begin{aligned} F(x, ay_1 + by_2) &= x^2 \frac{d}{dx}(ay_1 + by_2) = x^2 \left(\frac{d}{dx}(ay_1) + \frac{d}{dx}(by_2) \right) \\ &= ax^2 \frac{dy_1}{dx} + bx^2 \frac{dy_2}{dx} \\ &= aF(x, y_1) + bF(x, y_2) \quad \forall a, b \implies \textit{linear in } y \end{aligned}$$

Classification of ODEs by Linearity

Example (Linear vs nonlinear)

Consider $F(x, y, y') = x^2 y \frac{dy}{dx}$. As previously, *nonlinear in x*

Is F linear wrt y ? Check:

$$\begin{aligned}
 F(x, ay_1 + by_2) &= x^2 (ay_1 + by_2) \frac{d}{dx} (ay_1 + by_2) \\
 &= x^2 (ay_1 + by_2) \left(\frac{d}{dx} (ay_1) + \frac{d}{dx} (by_2) \right) \\
 &= x^2 (ay_1 + by_2) a \frac{dy_1}{dx} + x^2 (ay_1 + by_2) b \frac{dy_2}{dx} \\
 &= a^2 x^2 y_1 \frac{dy_1}{dx} + b^2 x^2 y_2 \frac{dy_2}{dx} + ab x^2 y_2 \frac{dy_1}{dx} + ab x^2 y_1 \frac{dy_2}{dx} \\
 &= a^2 F(x, y_1) + b^2 F(x, y_2) + ab x^2 y_2 \frac{dy_1}{dx} + ab x^2 y_1 \frac{dy_2}{dx} \\
 &\neq a F(x, y_1) + b F(x, y_2) \quad \forall a, b \quad \implies \text{nonlinear in } y
 \end{aligned}$$

Classification of ODEs by Linearity

Example (Linear vs nonlinear)

Consider $F(x, y, y') = x^2 \frac{dy}{dx} + e^x$.

Is F linear wrt x ? NO, as previously. $\implies$ *nonlinear in x*

Is F linear wrt y ?

$$\begin{aligned}
 F(x, ay_1 + by_2) &= x^2 \frac{d}{dx} (ay_1 + by_2) + e^x \\
 &= x^2 \left(\frac{d}{dx} (ay_1) + \frac{d}{dx} (by_2) \right) + e^x \\
 &= ax^2 \frac{dy_1}{dx} + bx^2 \frac{dy_2}{dx} + e^x \\
 &= a \left(x^2 \frac{dy_1}{dx} + e^x \right) + b \left(x^2 \frac{dy_2}{dx} + e^x \right) + (1 - a - b)e^x \\
 &= a F(x, y_1) + b F(x, y_2) + (1 - a - b)e^x \\
 &\neq a F(x, y_1) + b F(x, y_2) \quad \forall a, b \quad \implies \text{nonlinear in } y
 \end{aligned}$$

Classification of ODEs by Linearity

Generalizing these examples, it seems that any *first order linear ODE* must be possible to write as

$$a_1(x) \frac{dy}{dx} + a_0(x) y = 0$$

where $a_1(x)$ and $a_0(x)$ *can be nonlinear functions of x* .

Generalizing further, it seems that any *n^{th} order linear ODE* must be possible to write as

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x) y = 0$$

where $a_n(x)$, $a_{n-1}(x)$, ..., $a_0(x)$, *can be nonlinear functions of x* .

What about non-zero functions $g(x)$ on the RHS?

Classification of ODEs by Linearity

Recall terminology for **linear algebraic equations**, e.g., solve for variables v_1 and v_2 in:

$$av_1 + bv_2 = b_1$$

$$cv_1 + dv_2 = b_2$$

Equivalently, solve for v_1 and v_2 in the matrix equation

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

i.e., solve for $\vec{v}$ in $A\vec{v} = \vec{b}$

Here, A represents a **linear transformation**, or **linear function**, whereas $A\vec{v} = \vec{b}$ is a **linear equation**.

Classification of ODEs by Linearity

Analogously, a first order ODE in the form

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad (1)$$

can be written

$$\left(a_1(x) \frac{d}{dx} + a_0(x) \right) y = g(x)$$

So, if we define

$$L = \left(a_1(x) \frac{d}{dx} + a_0(x) \right)$$

then we can write Equation (1) as

$$L[y] = g(x)$$

Here, L is said to be a **linear operator** because

$$L[ay_1 + by_2] = a L[y_1] + b L[y_2]$$

whereas $L[y] = g(x)$ is said to be a **first order linear ODE**.

Classification of ODEs by Linearity

In the linear ODE

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

we say the equation is *homogeneous* if and only if $g(x) \equiv 0$.

Otherwise, we say the ODE is *nonhomogeneous*, or *inhomogeneous*.

Poll

- Go to
https://www.childsmath.ca/childsa/forms/main_login.php
- Click on **Math 2C03**
- Click on **Take Class Poll**
- Fill in poll **Linear and homogeneous ODEs**
- .

Classification of ODEs by Linearity

Example (Linear? Homogeneous?)

$$x^2 \frac{dy}{dx} + e^x y = 0$$

linear, homogeneous

$$x^2 \frac{dy}{dx} + e^y y = 0$$

nonlinear, homogeneous

$$(x^2 + e^x y) \frac{dy}{dx} = 3 \sin x$$

nonlinear, nonhomogeneous

$$(x^2 + e^x) \frac{dy}{dx} = 3 \sin x^2$$

linear, nonhomogeneous

Solving linear 1st order ODEs

How can we *solve* a first order linear ODE?

Solving linear 1st order ODEs

The general linear first order ODE is

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x) \quad (2)$$

where $a_1(x)$ and $a_0(x)$ *can be nonlinear functions of x* .

If $a_1(x) \neq 0$, we define

$$P(x) = \frac{a_0(x)}{a_1(x)}, \quad f(x) = \frac{g(x)}{a_1(x)}$$

and we can then write Equation (2) as

$$\frac{dy}{dx} + P(x)y = f(x) \quad (3)$$

This is the *standard form* of a first order *linear* ODE.

Solving linear 1st order ODEs

Recall: The *normal form* of Equation (3) is

$$\frac{dy}{dx} = -P(x)y + f(x) \quad (4)$$

What condition(s) will guarantee a unique solution?

Sufficient for existence and uniqueness of solution:

$P(x)$ and $f(x)$ continuous on an interval I

Why is this sufficient?

- RHS of Equation (4) is continuous at any point (x, y) where $P(x)$ and $f(x)$ are continuous wrt x ;
- $\frac{\partial}{\partial y}(\text{RHS Eq. (4)}) = -P(x)$ is continuous at any point (x, y) where $P(x)$ is continuous wrt x .

Solving linear 1st order ODEs

∴ We seek solutions of Equation (3),

$$\frac{dy}{dx} + P(x)y = f(x)$$

on an interval I where $P(x)$ and $f(x)$ are continuous.

Let's first solve the special case with $f(x) \equiv 0$:

$$\frac{dy}{dx} + P(x)y = 0$$

This homogeneous equation is *separable*...

Solving linear 1st order ODEs

Homogeneous first order linear equation: $\frac{dy}{dx} + P(x)y = 0$

- One solution is $y(x) \equiv 0$.
- No other solution can pass through $y = 0$
(by existence-uniqueness theorem), so now assume $y \neq 0$.

$$\begin{aligned}\frac{1}{y} \frac{dy}{dx} = -P(x) &\implies \int \frac{dy}{y} = - \int P(x) dx \\ &\implies \ln |y| + C = - \int P(x) dx \\ &\implies |y| = e^{-C} e^{-\int P(x) dx} \\ &\implies y = c e^{-\int P(x) dx} \quad c \in \mathbb{R}\end{aligned}$$

Note: For any constant C , $e^{-C} > 0$, but we can drop the absolute value bars on y by replacing e^{-C} with arbitrary $c \in \mathbb{R}$.

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation:

$$\frac{dy}{dx} + P(x)y = f(x)$$

Critical insight:

If $y_p(x)$ is a **particular solution** of the nonhomogeneous equation and $y_c(x) = c e^{-\int P(x) dx}$ is any solution of the homogeneous equation, then $y_c + y_p$ is another particular solution of the nonhomogeneous equation. Check:

$$\begin{aligned}\frac{d}{dx}(y_c + y_p) + P(x)(y_c + y_p) &= \left(\frac{dy_c}{dx} + P(x)y_c\right) + \left(\frac{dy_p}{dx} + P(x)y_p\right) \\ &= 0 + f(x) \\ &= f(x)\end{aligned}$$

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation:

$$\frac{dy}{dx} + P(x)y = f(x)$$

Stroke of genius allows us to find a particular solution:

The parameter c in $y_c(x) = c e^{-\int P(x) dx}$ is a *constant*.

What if we allow c to vary as a function of x ?...

Can we get $f(x)$ rather than 0 on the RHS?

Try plugging $y = u(x)y_1(x)$ into the nonhomogeneous equation, and see if we can solve for $u(x)$...

Note: This idea is known as *variation of parameters*.

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation:

$$\frac{dy}{dx} + P(x)y = f(x)$$

Try $y = u(x)y_1(x) = u(x)e^{-\int P(x) dx}$

$$\begin{aligned}\frac{d(uy_1)}{dx} + P(x)(uy_1) &= u\frac{dy_1}{dx} + y_1\frac{du}{dx} + P(x)uy_1 \\ &= y_1\frac{du}{dx} + u\left(\frac{dy_1}{dx} + P(x)y_1\right) \\ &= \left(y_1\frac{du}{dx}\right) + 0\end{aligned}$$

So we need $y_1\frac{du}{dx} = f(x)$

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation:

$$\frac{dy}{dx} + P(x)y = f(x)$$

$$y_1(x) \frac{du}{dx} = f(x) \implies \frac{du}{dx} = \frac{f(x)}{y_1(x)} \implies du = \frac{f(x)}{y_1(x)} dx$$

$$\implies u = \int \frac{f(x)}{y_1(x)} dx = \int f(x) e^{\int P(x) dx} dx$$

$$\implies y_p = uy_1 = e^{-\int P(x) dx} \int f(x) e^{\int P(x) dx} dx$$

$$\implies y = y_c + y_p = c e^{-\int P(x) dx} + e^{-\int P(x) dx} \int f(x) e^{\int P(x) dx} dx$$

We call $e^{\int P(x) dx}$ an **integrating factor**.

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation in standard form:

$$\frac{dy}{dx} + P(x)y = f(x) \quad (*)$$

- Integrating factor: $e^{\int P(x) dx}$
- Recall: $\frac{d}{dx} \left(e^{\int P(x) dx} \right) = e^{\int P(x) dx} \cdot f'(x)$ (*chain rule*)
- Recall: $\frac{d}{dx} \int f(x) dx = f(x)$ (*defn of antiderivative*)
- Derivative of integrating factor:

$$\begin{aligned} \frac{d}{dx} \left(e^{\int P(x) dx} \right) &= e^{\int P(x) dx} \frac{d}{dx} \left(\int P(x) dx \right) \\ &= e^{\int P(x) dx} P(x) \end{aligned}$$

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation in standard form:

$$\frac{dy}{dx} + P(x)y = f(x) \quad (*)$$

- Consequently, derivative of “(integrating factor) $\times$ y ” is, from the product rule:

$$\begin{aligned} \frac{d}{dx} \left(e^{\int P(x) dx} y \right) &= e^{\int P(x) dx} \frac{dy}{dx} + \left(e^{\int P(x) dx} P(x) \right) y \\ &= e^{\int P(x) dx} \underbrace{\left(\frac{dy}{dx} + P(x)y \right)}_{\text{LHS of } (*)} \end{aligned}$$

- Note that integrating a derivative is trivial:

$$\int \frac{d}{dx} \left(e^{\int P(x) dx} y \right) dx = \left(e^{\int P(x) dx} y \right) + \text{constant}$$

Solving linear 1st order ODEs

Nonhomogeneous first order linear equation in standard form:

$$\frac{dy}{dx} + P(x)y = f(x) \quad (*)$$

EMSID *method to find the general solution:*

- 1 **E**valuate the integrating factor $e^{\int P(x) dx}$
- 2 **M**ultiply both sides of $(*)$ by the integrating factor
- 3 **S**implify using product rule:

$$\frac{d}{dx} \left(e^{\int P(x) dx} y \right) = e^{\int P(x) dx} f(x) \quad (**)$$

- 4 **I**ntegrate both sides of $(**)$ (but add only one constant).
- 5 **D**ivide both sides by the integrating factor (to isolate y).

Note: "EMSID" isn't standard terminology; it's just a suggestion to help you remember the method.

Solving linear 1st order ODEs

Example (Find the general solution of $\frac{dy}{dx} = x - y$)

Convert to standard form: $\frac{dy}{dx} + y = x$ (♠)
 $P(x) \equiv 1$ and $f(x) = x$.

Evaluate the integrating factor: $e^{\int P(x)dx} = e^{\int dx} = e^x$

Multiply (♠) by e^x : $e^x \left(\frac{dy}{dx} + y \right) = e^x x$

Simplify using product rule: $\frac{d}{dx} (e^x y) = x e^x$

Integrate: $e^x y = \int x e^x dx$ *integrate by parts*
 $= (x - 1)e^x + c, \quad c \in \mathbb{R}$

Divide: $y = (x - 1) + c e^{-x}, \quad c \in \mathbb{R}$

Solving linear 1st order ODEs

Example (Find the general solution of $\frac{dy}{dt} = e^{-t^2} \cos(t) - 2ty$)

Convert to standard form: $\frac{dy}{dt} + 2ty = e^{-t^2} \cos(t) \quad (\spadesuit)$
 $P(t) = 2t \text{ and } f(t) = e^{-t^2} \cos(t).$

Evaluate the integrating factor: $e^{\int P(t)dt} = e^{\int 2t dt} = e^{t^2}$

Multiply $(\spadesuit)$ by e^{t^2} : $e^{t^2} \left(\frac{dy}{dt} + 2ty \right) = e^{t^2} e^{-t^2} \cos(t) = \cos(t)$

Simplify using product rule: $\frac{d}{dt} (e^{t^2} y) = \cos(t)$

Integrate: $e^{t^2} y = \int \cos(t) dt = \sin(t) + c, \quad c \in \mathbb{R}$

Divide: $y = e^{-t^2} (\sin(t) + c), \quad c \in \mathbb{R}$

Linear ODE Models

Population models we've discussed

Malthusian model: $\frac{dP}{dt} = r P$

■ Linear? *YES.*

■ Standard form? $\frac{dP}{dt} - r P = 0$ (♠)

■ **BEWARE:** The dependent variable here is P , so we cannot use the usual notation $P(t)$ for the coefficient $(-r)$ of P on the LHS of the standard form (♠) !!

■ Integrating factor? $e^{\int -r dt} = e^{-rt}$

■ For this equation, EMSID works but is more cumbersome than just separating and integrating.

Population models we've discussed

Logistic model:
$$\frac{dP}{dt} = r P \left(1 - \frac{P}{K}\right)$$

- Linear? *NO.*
- Can we rewrite the equation somehow so that it becomes linear?
- What if we swap the roles of the independent and dependent variables?

$$\frac{dt}{dP} = \frac{1}{r P \left(1 - \frac{P}{K}\right)}$$

This equation is linear in standard form. We can solve it for the function $t(P)$ (and then hopefully invert to get $P(t)$).

- Good exercise: solve the logistic equation via EMSID method.

Note: EMSID works, but treating the logistic model as a separable equation is more natural.

Newton's Law of Cooling

Imagine dropping a golf ball into a tub of ice water.

How does the temperature of the golf ball change with time?

Let T_m be the temperature of the medium (ice water).

Let T be the temperature of the golf ball.

Newton proposed, for situations like this, that

$$\frac{dT}{dt} = -k(T - T_m)$$

for some *cooling rate constant* $k > 0$ (with units $\frac{1}{\text{time}}$).

Newton's Law of Cooling

Natural question: If the golf ball initially has temperature $T_0 > T_m$, how long will it take to cool to the temperature T_m of the medium?

We should be able to answer this question if we can solve the IVP:

Solve $\frac{dT}{dt} = -k(T - T_m)$ *subject to* $T(0) = T_0 > T_m$.

This equation is linear and separable. It is easier to solve by separating. Since $T_0 \neq T_m$, we can divide by $(T - T_m)$:

$$\begin{aligned}\frac{dT}{dt} = -k(T - T_m) &\implies \int_0^t \frac{1}{(T - T_m)} \frac{dT}{dt} dt = - \int_0^t k dt \\ &\implies \int_{T(0)}^{T(t)} \frac{dT}{T - T_m} = -k(t - 0)\end{aligned}$$

Newton's Law of Cooling

$$\begin{aligned}\Rightarrow -kt &= \int_{T_0}^T \frac{dT}{T - T_m} = \ln |T - T_m| \Big|_{T_0}^T \\&= \ln |T - T_m| - \ln |T_0 - T_m| \\&= \ln \left| \frac{T - T_m}{T_0 - T_m} \right| \\&= \ln \left(\frac{T - T_m}{T_0 - T_m} \right) \quad \because T_0 > T \geq T_m\end{aligned}$$

$$\begin{aligned}\Rightarrow t &= -\frac{1}{k} \ln \left(\frac{T - T_m}{T_0 - T_m} \right) \\&= \frac{1}{k} \ln \left(\frac{T_0 - T_m}{T - T_m} \right)\end{aligned}$$

Newton's Law of Cooling

We have found:
$$t = \frac{1}{k} \ln \left(\frac{T_0 - T_m}{T - T_m} \right)$$

t is the time it takes to cool from initial temperature T_0 to temperature T .

How long will it take to cool to T_m ?

Forever! i.e., $t \rightarrow \infty$ as $T \rightarrow T_m$

But T gets close to T_m quickly.

What is “close”? and what is “quickly”?

What if temperature can be measured only to within some error tolerance ε ? How long will it take to cool from T_0 to $T_m + \varepsilon$?

Practice problems: linear first order equations

In [Trench, §2.1](#):

- Homogeneous equations and IVPs: Exercises 1–11
- General solutions and direction fields: Exercises 12–24
- IVPs and solution graphs: Exercises 25–37
- Answers involving definite integrals: Exercises 38–42
- A linear model: Exercise 43
- The principle of superposition: Exercise 46